

Database Technologies – Assignment 8

1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
D6_96342_Jagdish@localhost> SELECT * FROM orders WHERE cnum = (SELECT cnum FROM customers WHERE cname = 'Cisneros');  
+-----+-----+-----+-----+-----+  
| onum | amt   | odate   | cnum | snum |  
+-----+-----+-----+-----+-----+  
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |  
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |  
+-----+-----+-----+-----+-----+  
2 rows in set (0.001 sec)
```

7. Extract all the orders of Motika.

```
D6_96342_Jagdish@localhost> SELECT * FROM orders WHERE snum = (SELECT snum FROM salespeople WHERE sname = 'Motika');  
+-----+-----+-----+-----+-----+  
| onum | amt   | odate   | cnum | snum |  
+-----+-----+-----+-----+-----+  
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |  
+-----+-----+-----+-----+-----+  
1 row in set (0.001 sec)
```

8. Find all the orders of salespeople servicing customers in London.

```
D6_96342_Jagdish@localhost> SELECT * FROM orders WHERE snum IN (SELECT snum FROM customers WHERE city = 'London');  
+-----+-----+-----+-----+-----+  
| onum | amt   | odate   | cnum | snum |  
+-----+-----+-----+-----+-----+  
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |  
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |  
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |  
+-----+-----+-----+-----+-----+  
3 rows in set (0.001 sec)
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
D6_96342_Jagdish@localhost> SELECT * FROM orders WHERE amt > ANY (SELECT amt FROM orders WHERE cnum IN (SELECT cnum FROM customers WHERE city = 'London'));
```

onum	amt	odate	cnum	snum
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3011	9891.88	1990-10-04	2006	1001

6 rows in set (0.001 sec)

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
D6_96342_Jagdish@localhost> SELECT * FROM customers WHERE rating >= ANY (SELECT rating FROM customers WHERE cnum IN (SELECT cnum FROM orders WHERE snum = (SELECT snum FROM salespeople WHERE sname = 'Serres')));
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.001 sec)

11. Select customers who have a greater rating than any other customer in Rome.

```
D6_96342_Jagdish@localhost> SELECT * FROM customers WHERE rating >= (SELECT MAX(rating) FROM customers WHERE cnum IN (SELECT cnum FROM customers WHERE city = 'Rome'));
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.001 sec)

12. Select all orders that had amounts that were greater than at least one of the orders from '1990-10-04'.

```
D6_96342_Jagdish@localhost> SELECT * FROM orders WHERE amt > ANY (SELECT amt FROM orders WHERE odate = "1990-10-04");
```

onum	amt	odate	cnum	snum
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002
3011	9891.88	1990-10-04	2006	1001

8 rows in set (0.001 sec)

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
D6_96342_Jagdish@localhost> SELECT * FROM orders WHERE amt < ANY (SELECT amt FROM orders WHERE cnum IN (SELECT cnum FROM customers WHERE city = "San Jose"));
```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3007	75.75	1990-10-04	2004	1002
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002

8 rows in set (0.001 sec)

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
D6_96342_Jagdish@localhost> SELECT o.amt, c.cname, c.rating FROM orders o LEFT JOIN customers c ON o.cnum = c.cnum WHERE o.amt > (SELECT AVG(amt) FROM orders);
```

amt	cname	rating
5160.45	Liu	200
4723.00	Clemens	100
9891.88	Clemens	100

3 rows in set (0.001 sec)

14. Select those customers whose rating are higher than every customer in London.

```
D6_96342_Jagdish@localhost> SELECT * FROM customers WHERE rating > (SELECT MAX(rating) FROM customers WHERE city = "London");
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.001 sec)

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
D6_96342_Jagdish@localhost> SELECT * FROM salespeople s WHERE s.city NOT IN (SELECT city FROM customers c WHERE c.snum = s.snum);
```

snum	sname	city	comm
1004	Motika	London	0.11
1007	Rifkin	Barcelona	0.15
1003	Axelrod	New York	0.10

3 rows in set (0.001 sec)

9. Find names and numbers of all salesperson who have more than one customer.

```
D6_96342_Jagdish@localhost> SELECT sname, snum FROM salespeople WHERE snum IN (SELECT snum FROM customers GROUP BY snum HAVING COUNT(cnum) > 1);
```

sname	snum
Peel	1001
Serres	1002

2 rows in set (0.001 sec)

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
D6_96342_Jagdish@localhost> SELECT s.snum, s.sname, s.city, COUNT(c.cnum) FROM salespeople s INNER JOIN customers c ON c.snum = s.snum WHERE s.snum IN (SELECT snum FROM customers GROUP BY snum HAVING COUNT(cnum) > 1) GROUP BY s.sname, s.snum, s.city;
```

snum	sname	city	COUNT(c.cnum)
1001	Peel	London	2
1002	Serres	San Jose	2

2 rows in set (0.001 sec)

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
D6_96342_Jagdish@localhost> SELECT snum, SUM(amt) FROM orders GROUP BY snum HAVING SUM(amt) > (SELECT MAX(amt) FROM orders);
```

snum	SUM(amt)
1001	15382.07

1 row in set (0.001 sec)